1. SYSTEM ARCHITECTURE & HIGH-LEVEL TOPOLOGY

RAPPER FRIEND is built as a zero-latency, client-authoritative Single Page Application (SPA) designed to run both within desktop browsers and embedded touchscreen LCARS terminals. The runtime environment prioritizes high-throughput phonemic parsing (sub-10ms response on 64-bar lyrics), Web Audio API real-time synthesis, and persistent client-side data isolation without telemetry.

SUBSYSTEM	TECHNOLOGY STACK	SPECIFICATION / RUNTIME ROLE
Core UI Framework	**React 19.0.1 + Vite 8.3.0**	Functional components, hooks, React.memo for bar renders.
Language & Typing	**TypeScript 7.0.2 (ESNext)**	Strict type safety across phonemes, LCARS eras, and beat presets.
Styling & Layout	**Tailwind CSS v4.3.3**	Custom LCARS-26 design token theme, viewport-constrained.
Audio Engine	**Native Web Audio API**	Real-time multi-track procedural beat synthesis + audio element routing.
Data Persistence	**IndexedDB + LocalStorage**	IndexedDB for custom audio Blobs; LocalStorage for phrases & custom can
3D Graphics Subsystem	**Three.js 0.186.1 + WebGL**	Force-directed 3D semantic graph with OrbitControls & glow shaders.
Linguistic Compilation	**Custom TypeScript AST**	CMU / Datamuse / Wiktionary phonemic engine with regex caching.

2. LCARS-26 DESIGN TOKENS & COLOR SCHEMES

The visual interface strictly models Michael Okuda's LCARS graphic system across 5 historical Starfleet eras, switchable dynamically with zero layout shift:

- **TNG (The Next Generation):** Primary: #FF9900 (Amber) | Secondary: #CC6699 (Mauve) | Accent: #FFCC00 (Gold) | Aux: #9999FF (Periwinkle)
- **DS9 (Deep Space Nine):** Primary: #FF6600 (Dark Orange) | Secondary: #FF9900 (Amber) | Accent: #CC6633 (Bronze) | Aux: #FFCC66 (Wh
- **VOYAGER (Delta Quadrant):** Primary: #3399CC (Ice Blue) | Secondary: #99CCFF (Sky Blue) | Accent: #FF9900 (Amber) | Aux: #6666CC (Navy)
- **NEMESIS (Late 24th Century):** Primary: #009999 (Teal) | Secondary: #33CCCC (Cyan) | Accent: #FF6633 (Coral) | Aux: #669999 (Slate)
- **DISCOVERY (32nd Century):** Primary: #00E5FF (Neon Cyan) | Secondary: #7C4DFF (Purple) | Accent: #FFD600 (Yellow) | Aux: #FF1744 (Red A

3. PROCEDURAL WEB AUDIO API SYNTHESIS ENGINE

The ProceduralBeatSynthesizer class (/src/utils/instrumentalBeatEngine.ts) generates 12 authentic regional instrumentals entirely via programmatic Web Audio synthesis, with zero external sample loading latency:

KICK & 808 SUB-BASS OSCILLATOR TOPOLOGY

Uses an OscillatorNode (sine wave) pitched at 145 Hz, executing an exponential frequency ramp down to 42–48 Hz over 90ms, continuing into a sub-bass decay tail to 32 Hz over 240–420ms. The GainNode applies a linear attack followed by an exponential decay to 0.001. In long 808 modes (Drill/Memphis), frequency glide between consecutive 16th steps is handled via exponentialRampToValueAtTime.

SNARE, CLAP & NOISE WIRE TOPOLOGY

Synthesizes snappy snare wires using an AudioBufferSourceNode filled with white noise (Math.random() * 2 - 1) passed through a BiquadFilterNode (highpass at 750–1400 Hz). The acoustic body is reinforced by a parallel triangle oscillator sweeping from 195 Hz down to 110 Hz in 80ms.

HI-HAT & ROLLING TRIPLET SEQUENCER

Generates crisp closed and open hi-hats with an exponential-decay noise buffer filtered by a 6.8 kHz highpass BiquadFilter. Triplet rolls (hat pattern code 3) schedule three successive micro-bursts at t, t + stepDur*0.33, and t + stepDur*0.66 within a single 16th-note step.

MPC SWING ALGORITHM & MICRO-TIMING MATHEMATICS

Calculates step duration in milliseconds: baseStepMs = (60 / BPM / 4) * 1000. For swung feels (e.g. 90 BPM Boom-Bap at 24% swing, UKG 2-Step at 26% swing), odd steps are delayed by +(baseStepMs * swingFactor) while even steps are compressed by -(baseStepMs * swingFactor).

4. CUSTOM BEAT UPLOADER & INDEXEDDB SCHEMA

User-uploaded beats are persisted in an IndexedDB database ("rapper_friend_instrumentals_db", store "custom_beats") to bypass the 5MB browser LocalStorage limit and enable multi-megabyte audio file storage:

```
interface CustomUploadedBeat {
  id: string;               // UUID format 'custom-beat-{timestamp}-{idx}'
  name: string;             // Sanitized filename without extension
  fileName: string;         // Original file name (e.g. 'grime_instrumental.wav')
  estimatedBpm: number;     // User-entered or default BPM
  styleTag: string;         // Genre tag for indexing
  createdAt: number;        // Epoch timestamp
  audioBlob: Blob;          // Raw audio binary (audio/mp3, audio/wav, audio/ogg)
}
```

5. PHONEMIC ANALYSIS & SYLLABLE COUNTING HEURISTICS

The lyric analysis engine (/src/utils/rhymeEngine.ts) compiles written bars into structured phonetic telemetry in sub-millisecond execution:

- **Syllable Counting Engine:** Uses a multi-tier heuristic: (1) Exact lookup against an embedded 300+ word phonetic database; (2) Diphthong and vowel group reduction (ia, io, ou, ee, oo, ai, ay, oy); (3) Silent terminal-e reduction (except words ending in "-le"); (4) Prefix/suffix adjustment (-ed, -es, -tion, -ment, -ing).
- **Rhyme Density Scoring Formula:** Calculates the ratio of rhyming vowel/consonant syllables to total syllables in the stanza: RhymeDensity = (RhymingSyllables / TotalSyllables) * 100. Stanzas exceeding 70% earn the S+ "HIGH DENSITY" rating.
- **Automated Rhyme Scheme Detection:** Evaluates normalized terminal phonemes across lines 1..N. Identifies strict AABB, alternating ABAB, monorhyme AAAA, and irregular multi-bar enjambment chains.
- **Literary Device Scanner:** Automated regex and phonetic AST traversal identifying: Alliteration (3+ identical onset consonants), Assonance (identical vowel sequences), Consonance, Anaphora (repetition of phrase onsets), Chiasmus (ABBA syntactic inversions), and Polysyllabic chains (3+ syllables).

6. MULTISYLLABIC VOWEL CHORD MATHEMATICAL ENGINE

In /src/components/multis/MultiRhymeEngine.tsx, multisyllabic rhyming is computed as an ordered sequence of vowel phonemes regardless of intervening consonants:

```
[Target Phrase] = "astrometric connections"
 V o w e l   S t r i n g      =  [ æ  -  o Š  -  [  -  j  -  Y  -  [  -  Y n ]
 M a t c h i n g   C h o r d s  =  " h o l o g r a p h i c  p r o j e c t i o n s "  [ R  -  o Š  -  æ  -  j
                              =  " a t m o s p h e r i c  r e f l e c t i o n s "  [ æ  -  o Š  -  [  -  j
                              =  " b a r o m e t r i c  d i r e c t i o n s "  [ æ  -  o Š  -  [  -  j  -

Algorithmic Score = VowelMatchWeight(0.65) + SyllableMatchWeight(0.20) + ConsonantAssonance(0.15)
```

7. DIVERGENT RHYME SYSTEMS ENGINEERING MATRIX

The RhymeEvolutionsLab component (/src/components/evolutions/RhymeEvolutionsLab.tsx) models 12 distinct regional disciplines with specific acoustic and timing parameters:

DISCIPLINE ID	TEMPO / SUBDIVISION	MICRO-TIMING POCKET	CORE PHONETIC SHIFT
uk-grime-140	**140 BPM (Half/Double)**	+8ms Ahead (Staccato)	Non-Rhotic + Glottal Stop /ʔ/
uk-drill-142	**142 BPM (3+3+2 Tresillo)**	Deadpan Caesura Pause	MLE Price-Vowel /aj/ → /aɐ/
uk-garage-134	**134 BPM (Swung 2-Step)**	64% Swung Shuffling Kick	Alveolar Double-Tonguing
uk-britcore-90	**90 BPM (Council Boom-Bap)**	Dub Bass Sway Drag	TRAP-BATH Split (/æ/ → /ɑɐ/)
afro-amapiano-112	**112 BPM (3-2 Clavé)**	Syncopated Clavé Bell	Tonal High-Low-High Melodic
dancehall-98	**98 BPM (Dembow Riddim)**	Off-Beat Rub-a-Dub	Patois Open-A + Falling Glide
us-ny-boombap-90	**90 BPM (MPC60 Swing)**	-20ms Behind the Snare	Compound Multisyllabic Chord
us-abstract-80	**80 BPM (Drumless Loop)**	Conversational Rubato	Mosaic Word Boundary Dissolve
us-memphis-138	**138 BPM (12/8 Triplet)**	Quantized Triplet Roll	Southern Monophthong /aj/ → /ɑɐ/
us-gfunk-94	**94 BPM (G-Funk Bounce)**	-35ms Laid-Back Drawl	California Fronted /uɐ/ & /oˠ/
us-chopper-148	**148 BPM (32nd Artillery)**	Laser Quantized (0ms)	Coronal-Velar Double-Tonguing
us-detroit-106	**106 BPM (Urgent Piano)**	Overlapping Bar-Line Run	Punch-In Conversational Slant

8. 16-STEP BEAT GRID DATA SPECIFICATION

Each discipline models every sixteenth-note subdivision across the 4/4 bar with drum triggers and vocal roles: KICK, SNARE, CLAP, 808-GLIDE, BASS-STAB, CLAVE, SETUP, INTERNAL, SKIP-BURST, CAESURA-PAUSE, END-RHYME, and RELOAD-PUNCH.

9. 3D WEBGL FORCE GRAPH ARCHITECTURE

The Graph3DView component renders lexical relationships in interactive Three.js 3D subspace:

- **Three.js Scene Graph:** A custom PerspectiveCamera (FOV 60, aspect ratio dynamic, near 0.1, far 1000) rendered with WebGLRenderer (antialias true, alpha true). Nodes are represented by SphereGeometry with MeshPhongMaterial and additive glowing Sprite point clouds.
- **Force Simulation Physics:** Nodes experience Coulombic repulsion (F = k * q1 * q2 / r^2) and Hooke-spring attraction (F = -k * (x - x0)) along shared rhyme family edges, computed per animation frame with velocity damping (0.88).
- **Orbit & Touch Controls:** Full 360-degree rotation, pinch-to-zoom, and Raycaster node picking with instantaneous LCARS chirp audio feedback on hover and selection.

10. UNIVERSAL SUBSPACE LEXICON CASCADE

In /src/utils/universalDictionaryEngine.ts, queries cascade through three resilient data tiers:

- **Tier 1 (Built-In Canon):** Immediate in-memory lookup across 301 hand-curated Arcana, Poetic, Astrometric, AAVE, MLE, and PIE root nodes (0ms latency).
- **Tier 2 (IndexedDB Custom Canon):** User-pinned entries stored in browser storage, surviving cache clears and page reloads.
- **Tier 3 (Universal REST Cascade):** Query fallback across Datamuse REST API (words?sp=...&md=dpfs) and Free Dictionary API (api.dictionaryapi.dev/api/v2/entries/en/...) with local Map LRU caching (200 entries max).

11. PERFORMANCE BENCHMARKS & BUILD PROFILE

• **Initial Bundle Size:**	685 KB (gzip: 182 KB)	Three.js + Lucide + React 19 tree-shaken
• **Bar Compilation Latency:**	< 8 milliseconds	Calculated synchronously on 16-bar verses
• **Web Audio Latency:**	< 5 milliseconds	Direct AudioContext hardware destination buffer
• **IndexedDB Read/Write:**	< 15 milliseconds	Non-blocking transactional I/O for audio Blobs
• **FPS Target:**	60 FPS stable	Canvas2D and Three.js optimized requestAnimationFrame